© Ahmad Esmaili, Robert F. Kelly

1-1 AND RECURSIVE RELATIONSHIPS

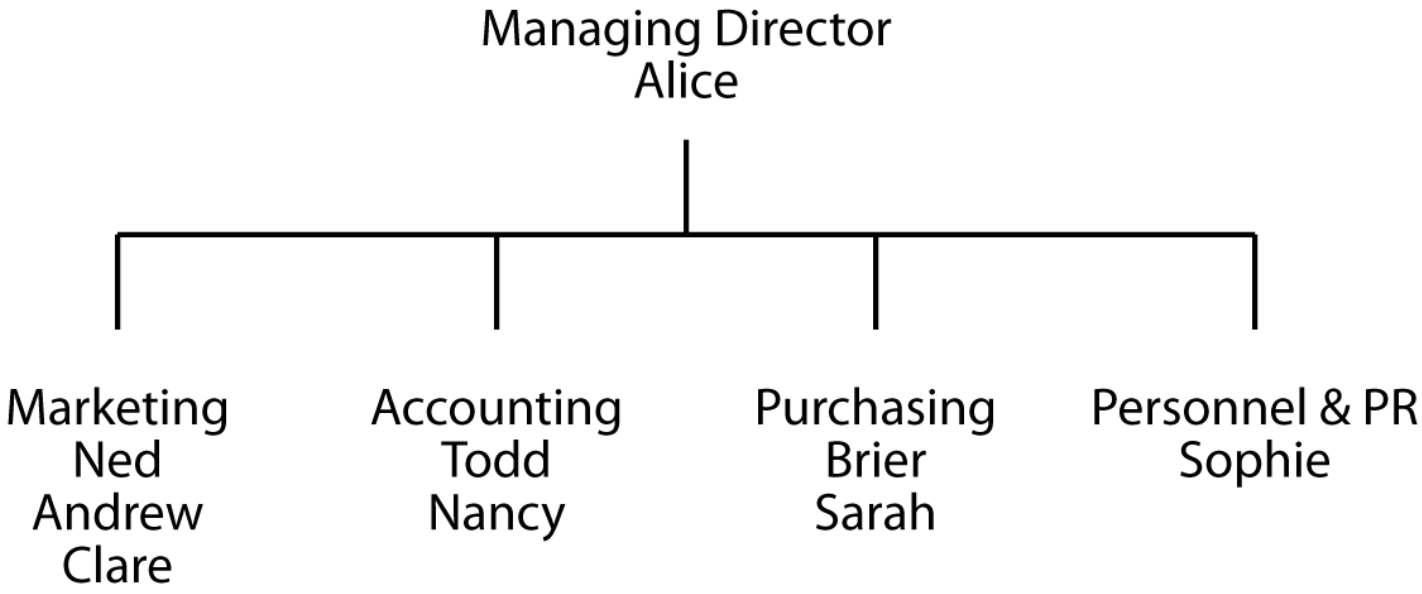
Reading: Chapter 6

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Organization Chart

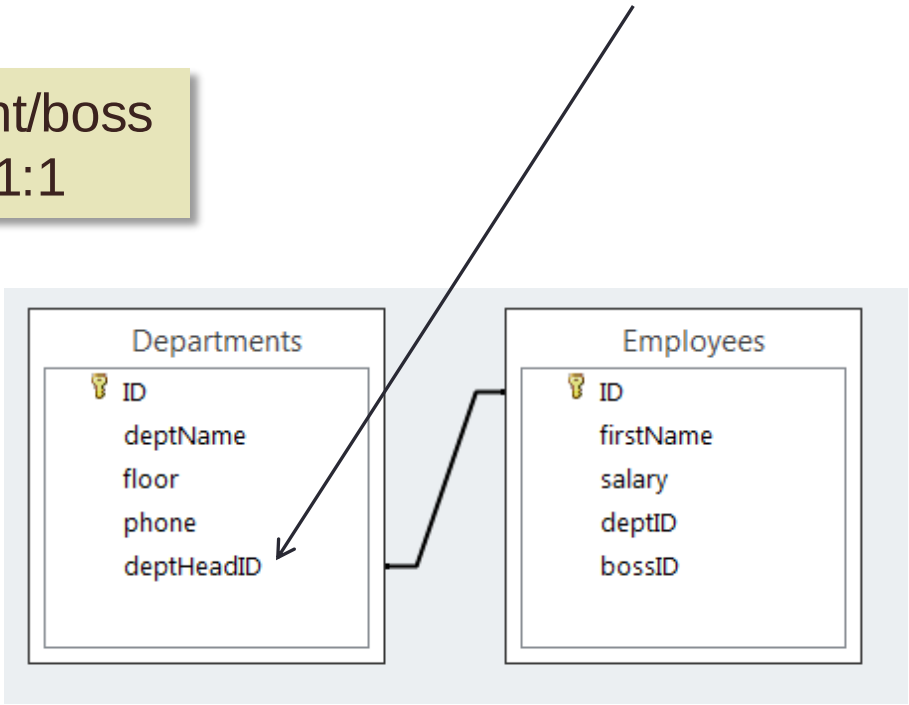
- Org charts implicitly show a relationship between employees (e.g., supervisor), but how do we show this in a DB



1-1 Relationships

- Org chart has 2 relationships
  - A department has one or more employees, but
  - Each department has only one department head

The department/boss relationship is 1:1

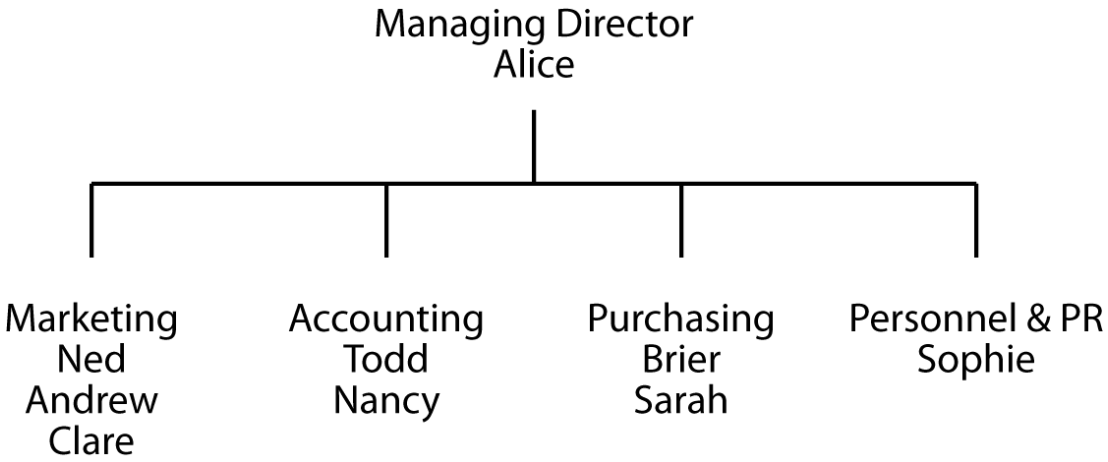


Mapping a 1:1 Relationship

- Usual rules apply (foreign key / primary key)
- Where do you put the foreign key?
  - Departments
  - Employees
  - Both tables

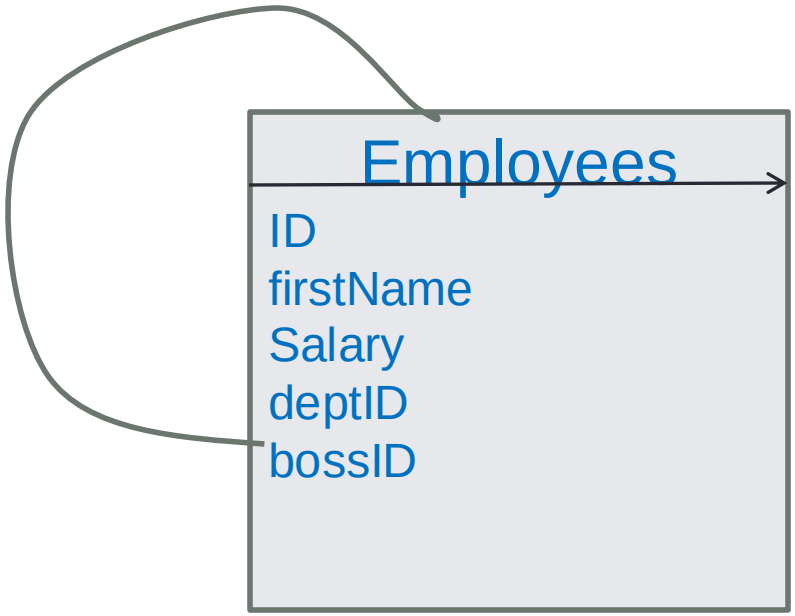
Recursive Relationships

- Example
  - Each employee has a boss and
  - Boss has many employees
- Both are in the same table (Employees)
- Define a recursive relationship



Modeling a Recursive Relationship

- A recursive relationship relates an entity to itself
- Label recursive relationships



Employees show a recursive relationship

Mapping a Recursive Relationship

- A recursive relationship relates an entity to itself

Departments | **Employees** | BossSalaryQuery

ID	firstName	salary	deptID	bossID
1	Alice	$75,000.00	1	
2	Ned	$45,000.00	2	1
3	Andrew	$25,000.00	2	2
4	Clare	$22,000.00	2	2
5	Todd	$38,000.00	3	3
6	Nancy	$22,000.00	3	5
7	Brier	$43,000.00	4	1
8	Sarah	$56,000.00	4	7
9	Sophie	$35,000.00	5	1

bossID is the foreign key to the ID primary key (same table)

Mapping a Recursive Relationship

- Usual rules
- 1:m
  - The entity gets an additional column for the foreign key
  - Need a name different from the primary key

Example

- Two relationships

Departments | Employees

ID	deptName	floor	phone	deptHeadID
1	Management	5	2001	1
2	Marketing	1	2002	2
3	Accounting	4	2003	5
4	Purchasing	4	2004	7
5	Personnel & PR	1	2005	9

1:1

recursive

Departments | Employees

ID	firstName	salary	deptID	bossID
1	Alice	$75,000.00	1	
2	Ned	$45,000.00	2	1
3	Andrew	$25,000.00	2	2
4	Clare	$22,000.00	2	2
5	Todd	$38,000.00	3	3
6	Nancy	$22,000.00	3	5
7	Brier	$43,000.00	4	1
8	Sarah	$56,000.00	4	7
9	Sophie	$35,000.00	5	1

Querying a 1:1 Relationship

- List the salary of each department's boss.

```
SELECT firstName, salary
FROM Employees
WHERE (Employees.ID IN
    (SELECT deptHeadID FROM Departments));
```

firstName	salary
Alice	$75,000.00
Ned	$45,000.00
Todd	$38,000.00
Brier	$43,000.00
Sophie	$35,000.00

Let's Look at a Sub-Query

- SELECT deptHeadID FROM Departments;

deptHeadID
1
2
5
7
9

Alternate SQL

- List the salary of each department's boss.

```
SELECT Employees.firstName, Departments.deptName,
Employees.salary
FROM Employees INNER JOIN Departments
    ON Employees.ID = Departments.deptHeadID;
```

Query11 | Relationships | Departments

firstName	deptName	salary
Alice	Management	$75,000.00
Ned	Marketing	$45,000.00
Todd	Accounting	$38,000.00
Brier	Purchasing	$43,000.00
Sophie	Personnel & PR	$35,000.00

Querying a Recursive Relationship

- Find the salary of Nancy's boss.

```
SELECT Employees.firstName, Employees.salary,
  Bosses.firstName, Bosses.salary, Employees.bossID
FROM Employees, Employees AS Bosses
WHERE (((Employees.firstName)='Nancy') AND
  ((Employees.bossID)=Bosses.ID));
```

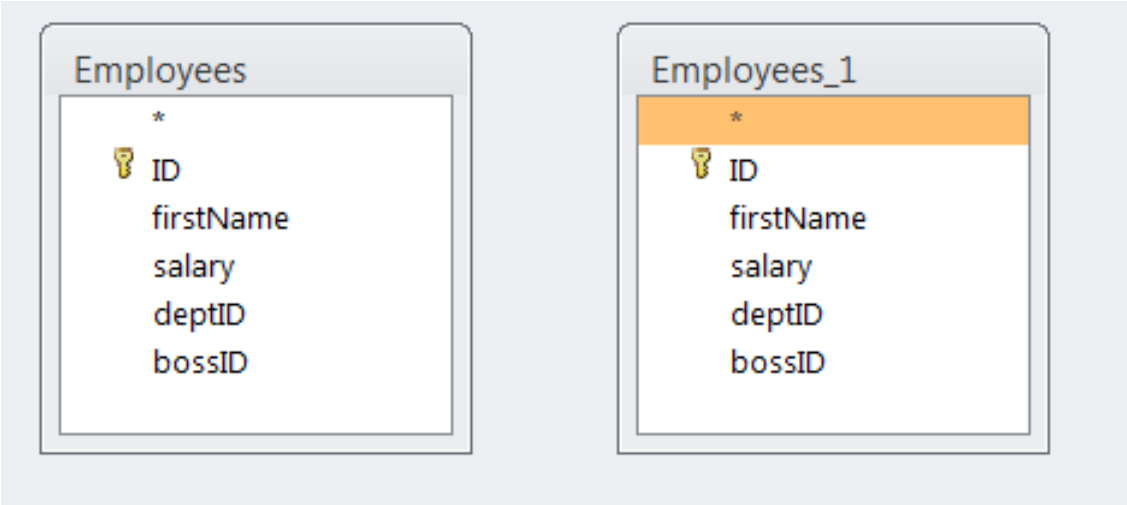
notice

Departments | Employees | **BossSalaryQuery**

Employees.firstName	Employees.salary	Bosses.firstName	Bosses.salary	bossID
Nancy	$22,000.00	Todd	$38,000.00	5

MS Access Design View

- Access will allow you to add multiple copies of a table to the design view
- Both tables will be available in the Field drop-down
- Will create a join of the copy with the original



Are We on Track?

- Write a query that will display the names of employees and their bosses

First build the new DB with the Excel data for the tables on Blackboard:

1. Employees.xlsx
2. Departments.xlsx

TrackBossName | Employees

Employee	Boss
Ned	Alice
Andrew	Ned
Clare	Ned
Todd	Andrew
Nancy	Todd
Brier	Alice
Sarah	Brier
Sophie	Alice
*	

Are We on Track

- Write a query that will display the department names and the name of each department head

TrackDeptBoss

Department	Department Head
Management	Alice
Marketing	Ned
Accounting	Todd
Purchasing	Brier
Personnel & PR	Sophie